

1. Write a Java program to display your name, age, and college name using variables of appropriate data types .

```
public class StudentDetails {  
    public static void main(String[] args) {  
        String name = "Arpit";  
        int age = 20;  
        String collegeName = "ABC College";  
  
        System.out.println("Name: " + name);  
        System.out.println("Age: " + age);  
        System.out.println("College Name: " + collegeName);  
    }  
}
```

Output- Name: Arpit kumar saini
Age: 19
College Name:Coer university

2. Input two integers and print the result of all arithmetic operations (addition, subtraction, multiplication, division, and modulus).

```
import java.util.Scanner;  
  
public class ArithmeticOperations {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter first integer: ");  
        int a = sc.nextInt();  
  
        System.out.print("Enter second integer: ");  
        int b = sc.nextInt();  
  
        System.out.println("Addition: " + (a + b));  
        System.out.println("Subtraction: " + (a - b));  
        System.out.println("Multiplication: " + (a * b));  
        System.out.println("Division: " + (a / b));  
        System.out.println("Modulus: " + (a % b));  
  
        sc.close();  
    }  
}
```

Output-Enter first integer: 20
Enter second integer: 6

Addition: 26
Subtraction: 14
Multiplication: 120
Division: 3
Modulus: 2

3. Write a Java program to check whether a given number is even or odd using the modulus operator and if-else

```
import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num % 2 == 0) {
            System.out.println(num + " is Even");
        } else {
            System.out.println(num + " is Odd");
        }

        sc.close();
    }
}
```

Output-Enter a number: 15
15 is Odd

4. Write a program to demonstrate the use of relational operators by comparing two numbers.

```
import java.util.Scanner;

public class RelationalOperators {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.println("a == b: " + (a == b));
        System.out.println("a != b: " + (a != b));
    }
}
```

```

        System.out.println("a > b: " + (a > b));
        System.out.println("a < b: " + (a < b));
        System.out.println("a >= b: " + (a >= b));
        System.out.println("a <= b: " + (a <= b));

        sc.close();
    }
}

```

Output- Enter first number: 10
Enter second number: 20

```

a == b: false
a != b: true
a > b: false
a < b: true
a >= b: false
a <= b: true

```

5. Accept a floating-point number and convert it explicitly to an integer. Print both the original and the converted values.

```

import java.util.Scanner;

public class FloatToInteger {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a floating-point number: ");
        double number = sc.nextDouble();

        int integerNumber = (int) number; // Explicit type casting

        System.out.println("Original value: " + number);
        System.out.println("Converted integer value: " + integerNumber);

        sc.close();
    }
}

```

Output-Enter a floating-point number: 45.67
Original value: 45.67
Converted integer value: 45

6. Write a program to accept a character and print its ASCII value .

```
import java.util.Scanner;

public class ASCIIValue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        int ascii = (int) ch;

        System.out.println("Character: " + ch);
        System.out.println("ASCII value: " + ascii);

        sc.close();
    }
}
```

Output-Enter a character: A
Character: A
ASCII value: 65

7. Input a number and check if it is divisible by both 3 and 5 using logical operators.

```
import java.util.Scanner;

public class DivisibleByThreeAndFive {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num % 3 == 0 && num % 5 == 0) {
            System.out.println(num + " is divisible by both 3 and 5.");
        } else {
            System.out.println(num + " is not divisible by both 3 and 5.");
        }

        sc.close();
    }
}
```

Output-Enter a number: 30
30 is divisible by both 3 and 5.

8. Input a character and check whether it is a vowel or a consonant using if-else.

```
import java.util.Scanner;

public class VowelConsonant {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
            System.out.println("Vowel");
        } else {
            System.out.println("Consonant");
        }

        sc.close();
    }
}
```

Output-Enter a character: E
Vowel

9. Input marks of three subjects, calculate total and percentage, and print pass/fail using if-else (passing criteria: $\geq 40\%$ in all subjects).

```
import java.util.Scanner;

public class MarksResult {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks of Subject 1: ");
        double sub1 = sc.nextDouble();

        System.out.print("Enter marks of Subject 2: ");
        double sub2 = sc.nextDouble();

        System.out.print("Enter marks of Subject 3: ");
```

```

double sub3 = sc.nextDouble();

double total = sub1 + sub2 + sub3;
double percentage = total / 3;

System.out.println("Total Marks = " + total);
System.out.println("Percentage = " + percentage + "%");

if (sub1 >= 40 && sub2 >= 40 && sub3 >= 40) {
    System.out.println("Result: Pass");
} else {
    System.out.println("Result: Fail");
}

sc.close();
}
}

```

Output- Enter marks of Subject 1: 70
Enter marks of Subject 2: 65
Enter marks of Subject 3: 80
Total Marks = 215.0
Percentage = 71.66666666666667%
Result: Pass

10. Write a program to find the greatest of two numbers using if-else.

```

import java.util.Scanner;

class GreatestOfTwo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter two numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();

        if (a > b) {
            System.out.println(a + " is the greatest.");
        } else if (b > a) {
            System.out.println(b + " is the greatest.");
        } else {
            System.out.println("Both numbers are equal.");
        }

        sc.close();
    }
}

```

Output-Enter two numbers: 25 40
40 is the greatest.

11. Input three numbers and print the largest using nested if-else.

```
import java.util.Scanner;

class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter three numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a > b) {
            if (a > c) {
                System.out.println(a + " is the largest.");
            } else {
                System.out.println(c + " is the largest.");
            }
        } else {
            if (b > c) {
                System.out.println(b + " is the largest.");
            } else {
                System.out.println(c + " is the largest.");
            }
        }

        sc.close();
    }
}
```

Output-Enter three numbers: 25 40 30
40 is the largest.

12. Accept a year and check whether it is a leap year using if-else .

```
import java.util.Scanner;

class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```

System.out.print("Enter a year: ");
int year = sc.nextInt();

if (year % 400 == 0) {
    System.out.println(year + " is a leap year.");
} else if (year % 100 == 0) {
    System.out.println(year + " is not a leap year.");
} else if (year % 4 == 0) {
    System.out.println(year + " is a leap year.");
} else {
    System.out.println(year + " is not a leap year.");
}

sc.close();
}
}

```

Output-Enter a year: 2024
2024 is a leap year.

13. Accept a number and check whether it is positive, negative, or zero using if-else.

```

import java.util.Scanner;

class CheckNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num > 0) {
            System.out.println("The number is positive.");
        } else if (num < 0) {
            System.out.println("The number is negative.");
        } else {
            System.out.println("The number is zero.");
        }

        sc.close();
    }
}

```

Output-Enter a number: -5
The number is negative.

14. Accept a character and check if it is a digit, uppercase letter, lowercase letter, or a special character using character classification and if-else .

```
import java.util.Scanner;

public class CharacterClassification {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if (Character.isDigit(ch)) {
            System.out.println("It is a digit.");
        } else if (Character.isUpperCase(ch)) {
            System.out.println("It is an uppercase letter.");
        } else if (Character.isLowerCase(ch)) {
            System.out.println("It is a lowercase letter.");
        } else {
            System.out.println("It is a special character.");
        }

        sc.close();
    }
}
```

Example

Input:

G

Output:

It is an uppercase letter.

15. Accept a 3-digit number and check whether it is an Armstrong number using arithmetic operators and if-else.

```
import java.util.Scanner;

public class ArmstrongNumber {
    public static void main(String[] args) {
```

```

Scanner sc = new Scanner(System.in);

System.out.print("Enter a 3-digit number: ");
int num = sc.nextInt();

int original = num;

int digit1 = num % 10;
num = num / 10;

int digit2 = num % 10;
num = num / 10;

int digit3 = num % 10;

int sum = (digit1 * digit1 * digit1)
          + (digit2 * digit2 * digit2)
          + (digit3 * digit3 * digit3);

if (sum == original) {
    System.out.println("It is an Armstrong number.");
} else {
    System.out.println("It is not an Armstrong
number.");
}

sc.close();
}
}

```

Example

Input:

153

Output:

It is an Armstrong number.

16. Input salary and calculate tax based on income slabs using if-else:

Income $\leq$ 250000: No tax

250001 to 500000: 5%

500001 to 1000000: 20%

Above 1000000: 30%

```

import java.util.Scanner;

public class IncomeTax {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter salary: ");
        double salary = sc.nextDouble();

        double tax;

        if (salary <= 250000) {
            tax = 0;
        } else if (salary <= 500000) {
            tax = salary * 0.05;
        } else if (salary <= 1000000) {
            tax = salary * 0.20;
        } else {
            tax = salary * 0.30;
        }

        System.out.println("Tax = " + tax);

        sc.close();
    }
}

```

Example

Input:

Enter salary: 600000

Output:

Tax = 120000.0

17. Accept an integer and check whether it is a palindrome using arithmetic operations and type casting.

```

import java.util.Scanner;

public class PalindromeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

        System.out.print("Enter an integer: ");
        int num = sc.nextInt();

        int original = num;
        int reverse = 0;

        while (num != 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
        }

        if (original == reverse) {
            System.out.println("It is a palindrome.");
        } else {
            System.out.println("It is not a palindrome.");
        }

        sc.close();
    }
}

```

Example

Input:

121

Output:

It is a palindrome.

18. Accept a number and print whether it is prime or not using if-else.

```

import java.util.Scanner;

public class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        boolean prime = true;
    }
}

```

```

        if (num <= 1) {
            prime = false;
        } else {
            for (int i = 2; i < num; i++) {
                if (num % i == 0) {
                    prime = false;
                    break;
                }
            }
        }

        if (prime) {
            System.out.println("It is a prime number.");
        } else {
            System.out.println("It is not a prime number.");
        }

        sc.close();
    }
}

```

Example

Input:

```
Enter a number: 17
```

Output:

```
It is a prime number.
```

19. Accept a character and use if-else to identify whether it's a valid operator (+, -, *, /) and perform the corresponding operation on two input numbers.

```

import java.util.Scanner;

public class OperatorCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();
    }
}

```

```

        System.out.print("Enter an operator (+, -, *, /): ");
        char op = sc.next().charAt(0);

        if (op == '+') {
            System.out.println("Result = " + (num1 + num2));
        } else if (op == '-') {
            System.out.println("Result = " + (num1 - num2));
        } else if (op == '*') {
            System.out.println("Result = " + (num1 * num2));
        } else if (op == '/') {
            if (num2 != 0) {
                System.out.println("Result = " + (num1 /
num2));
            } else {
                System.out.println("Cannot divide by zero.");
            }
        } else {
            System.out.println("Invalid operator.");
        }

        sc.close();
    }
}

```

Example

Input:

```

Enter first number: 20
Enter second number: 5
Enter an operator (+, -, *, /): *

```

Output:

```

Result = 100.0

```

20. Accept age and gender and determine eligibility for marriage (male ≥ 21 , female ≥ 18) using nested if-else conditions.

```

import java.util.Scanner;

public class MarriageEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter age: ");
    }
}

```

```

        int age = sc.nextInt();

        System.out.print("Enter gender (M/F): ");
        char gender = sc.next().charAt(0);

        if (gender == 'M' || gender == 'm') {
            if (age >= 21) {
                System.out.println("Eligible for marriage.");
            } else {
                System.out.println("Not eligible for
marriage.");
            }
        } else if (gender == 'F' || gender == 'f') {
            if (age >= 18) {
                System.out.println("Eligible for marriage.");
            } else {
                System.out.println("Not eligible for
marriage.");
            }
        } else {
            System.out.println("Invalid gender.");
        }

        sc.close();
    }
}

```

Example

Input:

Enter age: 22

Enter gender (M/F): M

Output:

Eligible for marriage.

21. Accept three angles and check whether they can form a valid triangle (sum of angles = 180°).

```

import java.util.Scanner;

public class ValidTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

        System.out.print("Enter first angle: ");
        int a = sc.nextInt();

        System.out.print("Enter second angle: ");
        int b = sc.nextInt();

        System.out.print("Enter third angle: ");
        int c = sc.nextInt();

        if (a > 0 && b > 0 && c > 0 && (a + b + c == 180)) {
            System.out.println("The angles can form a valid
triangle.");
        } else {
            System.out.println("The angles cannot form a valid
triangle.");
        }

        sc.close();
    }
}

```

Example

Input:

```

Enter first angle: 60
Enter second angle: 60
Enter third angle: 60

```

Output:

```

The angles can form a valid triangle.

```

22. Input the length and breadth of a rectangle, and determine whether it is a square using if-else.

```

import java.util.Scanner;

public class SquareCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter length: ");
        int length = sc.nextInt();
    }
}

```



```

        System.out.print("Enter breadth: ");
        int breadth = sc.nextInt();

        if (length == breadth) {
            System.out.println("It is a square.");
        } else {
            System.out.println("It is not a square.");
        }

        sc.close();
    }
}

```

Example

Input:

```

Enter length: 10
Enter breadth: 10

```

Output:

```

It is a square.

```

23. Input a temperature in Celsius and convert it to Fahrenheit using type casting and expressions.

```

import java.util.Scanner;

public class TemperatureConversion {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();

        double fahrenheit = (celsius * 9 / 5) + 32;

        System.out.println("Temperature in Fahrenheit = " +
fahrenheit);

        sc.close();
    }
}

```

Example

Input:

```
Enter temperature in Celsius: 25
```

Output:

```
Temperature in Fahrenheit = 77.0
```

24. Accept a number and check whether it is a perfect number (sum of divisors = number) using basic arithmetic and if-else .

```
import java.util.Scanner;

public class PerfectNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int sum = 0;

        for (int i = 1; i < num; i++) {
            if (num % i == 0) {
                sum = sum + i;
            }
        }

        if (sum == num) {
            System.out.println("It is a perfect number.");
        } else {
            System.out.println("It is not a perfect number.");
        }

        sc.close();
    }
}
```

Example

Input:

```
Enter a number: 6
```

Output:

```
It is a perfect number.
```

25. Accept two integers and a logical operator as a character. Use if-else to perform the logical operation and display the result.

```
import java.util.Scanner;

public class LogicalOperation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int a = sc.nextInt();

        System.out.print("Enter second integer: ");
        int b = sc.nextInt();

        System.out.print("Enter logical operator (&, |, ^): ");
        char op = sc.next().charAt(0);

        int result;

        if (op == '&') {
            result = a & b;
            System.out.println("Result = " + result);
        } else if (op == '|') {
            result = a | b;
            System.out.println("Result = " + result);
        } else if (op == '^') {
            result = a ^ b;
            System.out.println("Result = " + result);
        } else {
            System.out.println("Invalid logical operator.");
        }

        sc.close();
    }
}
```

Example

Input:

```
Enter first integer: 5
Enter second integer: 3
Enter logical operator (&, |, ^): &
```

Output:

```
Result = 1
```

26. Input a five-digit number and check whether the first and last digits are the same using arithmetic expressions.

```
import java.util.Scanner;

public class FirstLastDigit {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a five-digit number: ");
        int num = sc.nextInt();

        int firstDigit = num / 10000;
        int lastDigit = num % 10;

        if (firstDigit == lastDigit) {
            System.out.println("First and last digits are the
same.");
        } else {
            System.out.println("First and last digits are not
the same.");
        }

        sc.close();
    }
}
```

Example

Input:

```
Enter a five-digit number: 12341
```

Output:

```
First and last digits are the same.
```

27. Input the cost price and selling price of an item. Determine profit or loss, and by how much, using if-else .

```
import java.util.Scanner;
```

```

public class ProfitLoss {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter cost price: ");
        double costPrice = sc.nextDouble();

        System.out.print("Enter selling price: ");
        double sellingPrice = sc.nextDouble();

        if (sellingPrice > costPrice) {
            double profit = sellingPrice - costPrice;
            System.out.println("Profit = " + profit);
        } else if (costPrice > sellingPrice) {
            double loss = costPrice - sellingPrice;
            System.out.println("Loss = " + loss);
        } else {
            System.out.println("No profit, no loss.");
        }

        sc.close();
    }
}

```

Example

Input:

```

Enter cost price: 500
Enter selling price: 650

```

Output:

```

Profit = 150.0

```

28. Accept a number and display the number of digits using expressions and if-else.

```

import java.util.Scanner;

public class NumberOfDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();
    }
}

```

```

        int n = Math.abs(num);

        if (n < 10) {
            System.out.println("Number of digits = 1");
        } else if (n < 100) {
            System.out.println("Number of digits = 2");
        } else if (n < 1000) {
            System.out.println("Number of digits = 3");
        } else if (n < 10000) {
            System.out.println("Number of digits = 4");
        } else if (n < 100000) {
            System.out.println("Number of digits = 5");
        } else {
            System.out.println("Number has more than 5
digits.");
        }

        sc.close();
    }
}

```

Example

Input:

```
Enter a number: 4567
```

Output:

```
Number of digits = 4
```

29. Input the height (in cm) of a person and classify them as short (<150), average (150–170), or tall (>170).

```

import java.util.Scanner;

public class HeightClassification {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter height in cm: ");
        double height = sc.nextDouble();

        if (height < 150) {
            System.out.println("Short");
        } else if (height <= 170) {

```

```

        System.out.println("Average");
    } else {
        System.out.println("Tall");
    }

    sc.close();
}
}

```

Example

Input:

Enter height in cm: 165

Output:

Average

30. Write a Java program to simulate a simple calculator that takes two numbers and an operator (+, -, *, /) and performs the corresponding operation using if-else.

```

import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        System.out.print("Enter operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        if (operator == '+') {
            System.out.println("Result = " + (num1 + num2));
        } else if (operator == '-') {
            System.out.println("Result = " + (num1 - num2));
        } else if (operator == '*') {
            System.out.println("Result = " + (num1 * num2));
        } else if (operator == '/') {
            if (num2 != 0) {

```

```
        System.out.println("Result = " + (num1 /
num2));
    } else {
        System.out.println("Cannot divide by zero.");
    }
} else {
    System.out.println("Invalid operator.");
}

    sc.close();
}
}
```

Example

Input:

```
Enter first number: 20
Enter second number: 4
Enter operator (+, -, *, /): /
```

Output:

```
Result = 5.0
```